Lecture 2

CHAPTER 1 FUNCTIONS:

1.1 PREVIEW OF CALCULUS

1.2 FUNCTIONS AND THEIR GRAPHS

1.3 COMBINING FUNCTIONS

1.4 TRIGONOMETRIC FUNCTIONS

1.5 EXPONENTIAL FUNCTIONS

1.4 TRIGONOMETRIC FUNCTIONS

Objectives

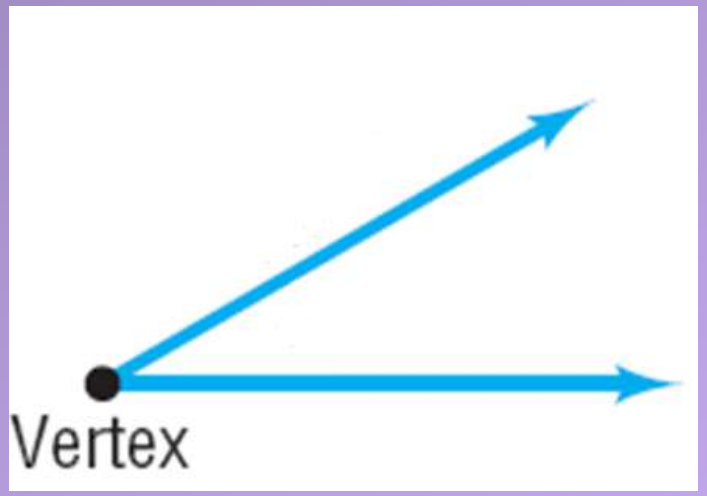
To review radian measure and the basic trigonometric functions.

Angles and Their Measure

Basic Terminology

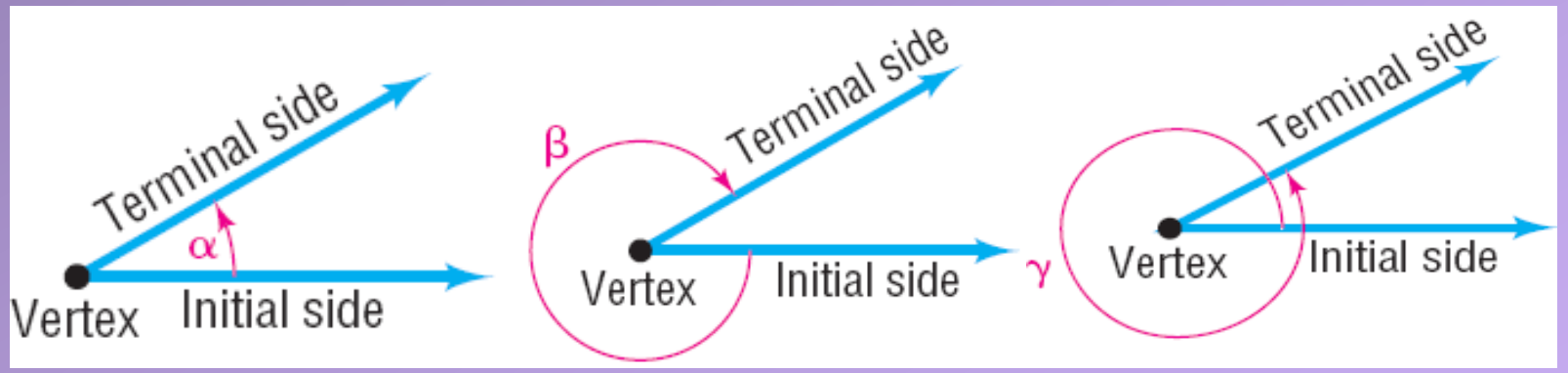
- Ray: A half-line starting at a vertex V

- Angle: Two rays with a common vertex



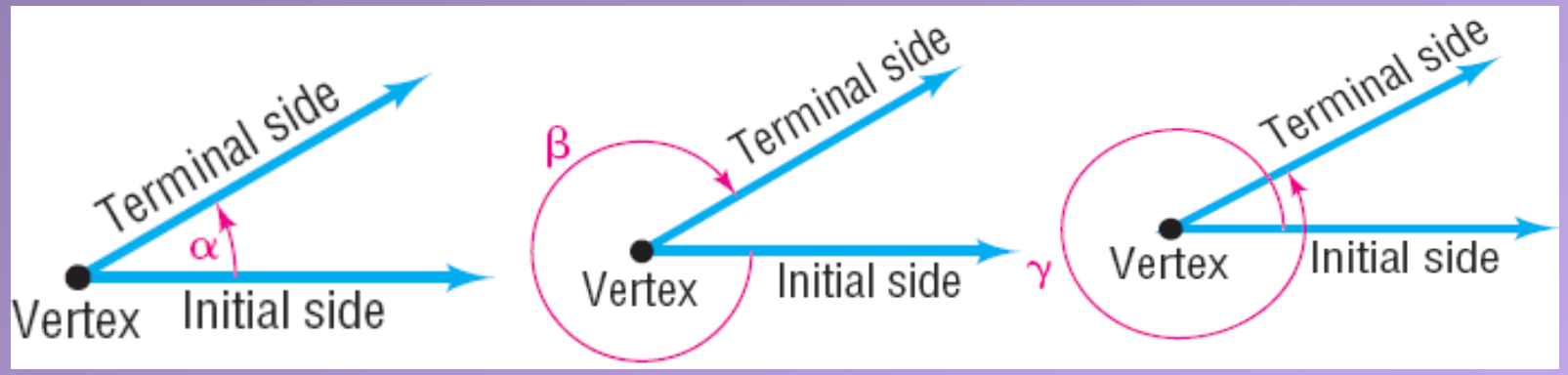
Basic Terminology

- Initial side and terminal side: The rays in an angle
  - Angle shows direction and amount of rotation
  - Lower-case Greek letters denote angles

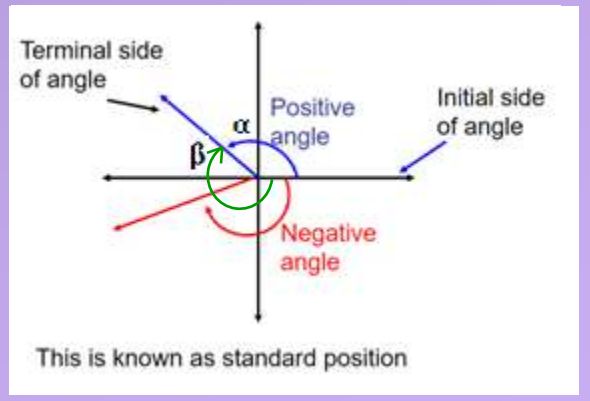


Basic Terminology

- Positive angle: Counterclockwise rotation
- Negative angle: Clockwise rotation
- Coterminal angles: Share initial and terminal sides (2 angles who have the same terminal sides)



OR



Positive angle **Negative angle** **Positive angle**

α, β = coterminal angles

To find positive or negative coterminal angles: add or subtract 360° or 2^x.

Measuring Angles

- Two usual ways of measuring
  - Degrees
    - 360° in one rotation
  - Radians
    - 2π radians in one rotation

360° = 2π radians

- Converting between radians and degrees
  - Convert degree to radian: 180º = π radians, 1º = $\frac{\pi}{180}$.

Ex. 270º = $270 \times \frac{\boldsymbol{\pi}}{\mathbf{180}} = \frac{3\pi}{2}$ **rad**.

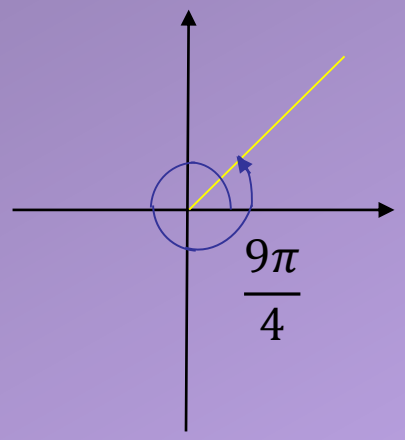
Measuring Angles

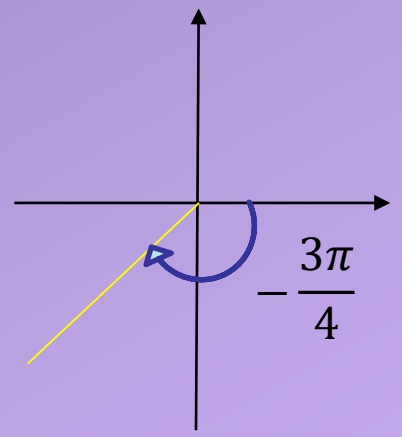
- Convert radian to degree:

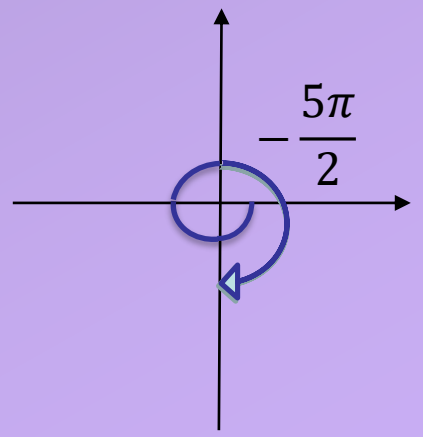
  π radians =180º, 1 rad = 180º/ π.

  Ex. $\frac{\pi}{6}$ rad = $\frac{\pi}{6}$ x $\frac{\mathbf{180^\circ}}{\boldsymbol{\pi}}$ = 30º; $-\frac{7\pi}{2} = -630º$; $2 = \frac{360^\circ}{\pi} \approx 114.59º$

- Nonzero radian measures can be positive or negative.







Radians vs. Degrees

- Measurements of common angles in degrees and radians

Degrees	$0°$	$30°$	$45°$	$60°$	$90°$	$120°$	$135°$	$150°$	$180°$
Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π
Degrees		$210°$	$225°$	$240°$	$270°$	$300°$	$315°$	$330°$	$360°$
Radians		$\frac{7\pi}{6}$	$\frac{5\pi}{4}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{7\pi}{4}$	$\frac{11\pi}{6}$	2π

- Supplementary angles are two angles that add up to 180 degrees or π radians.

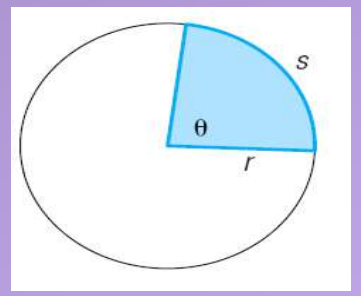
If you have an angle that measures 100°, then its supplementary angle can only measure 80°.

- Complimentary angles are two angles that add up to 90 degrees or $\pi/2$ radians.

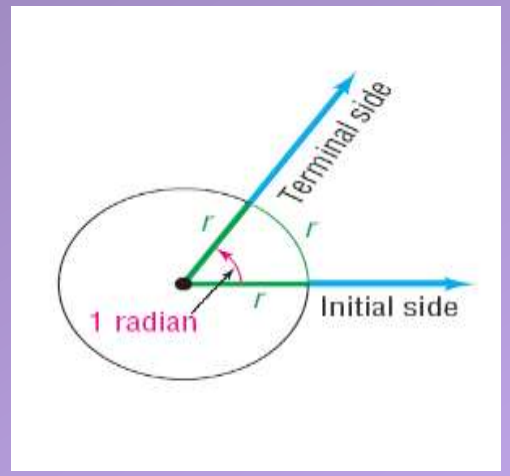
If you have an angle that measures 60°, then its complimentary angle can only measure 30°.

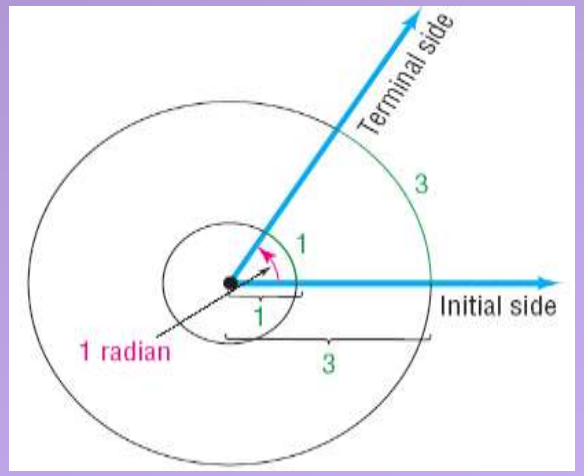
Radians

- Central angle: An angle whose vertex is at the center of a circle
  - Central angles subtend an arc, *s,* on the circle



- One radian is the measure of an angle which subtends an arc with length equal to the radius of the circle



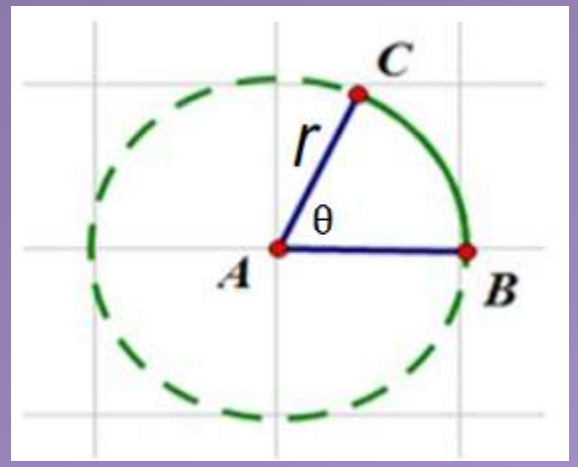


Radians

IMPORTANT!

- Radians are dimensionless
- If an angle appears with no units, it <u>must</u> be assumed to be in radians

Area of a Circular Sector



$$A = \frac{1}{2} r^2 \theta$$

A = sector area
r = radius
θ = angle measure (rad)

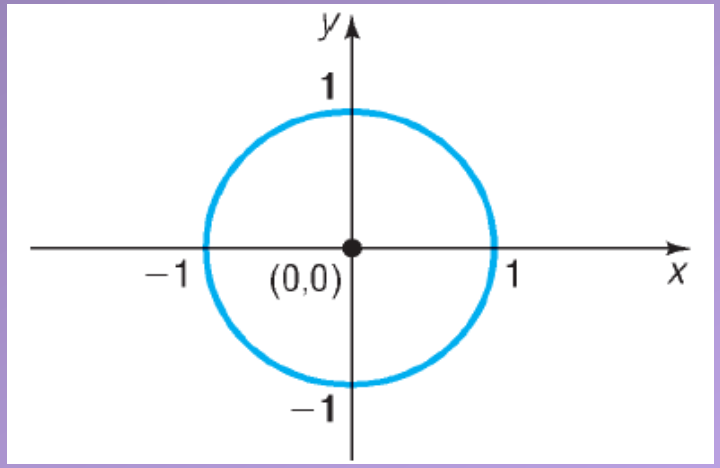
eg. Radius = 10 cm, area = 70 cm^2. Find the central angle of the sector.

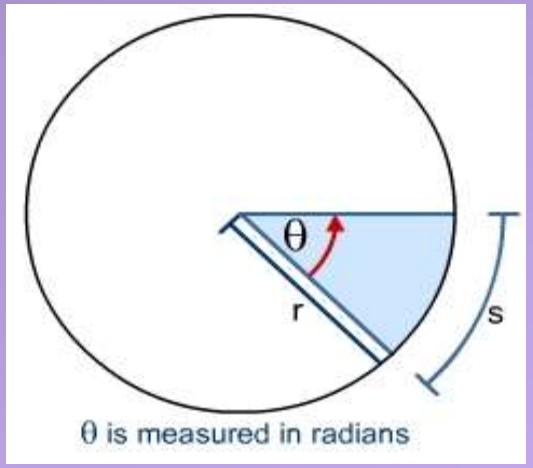
$A = \frac{1}{2} r^2 \theta$ ➔ 70 = $\frac{1}{2}(10^2)\theta$ ➔ $70 = 50\theta$

$\theta = \frac{70}{50} = \frac{7}{5}$ = 1.4 radian = $1.4(\frac{180}{\pi}) \approx 80.21°$

Unit Circle

- Unit circle: Circle with radius 1 centered at the origin
  - Equation: $x^2 + y^2 = 1$
  - Circumference: $2\pi r = 2\pi(1) = 2\pi$
  - Arc length: $s = r\theta$



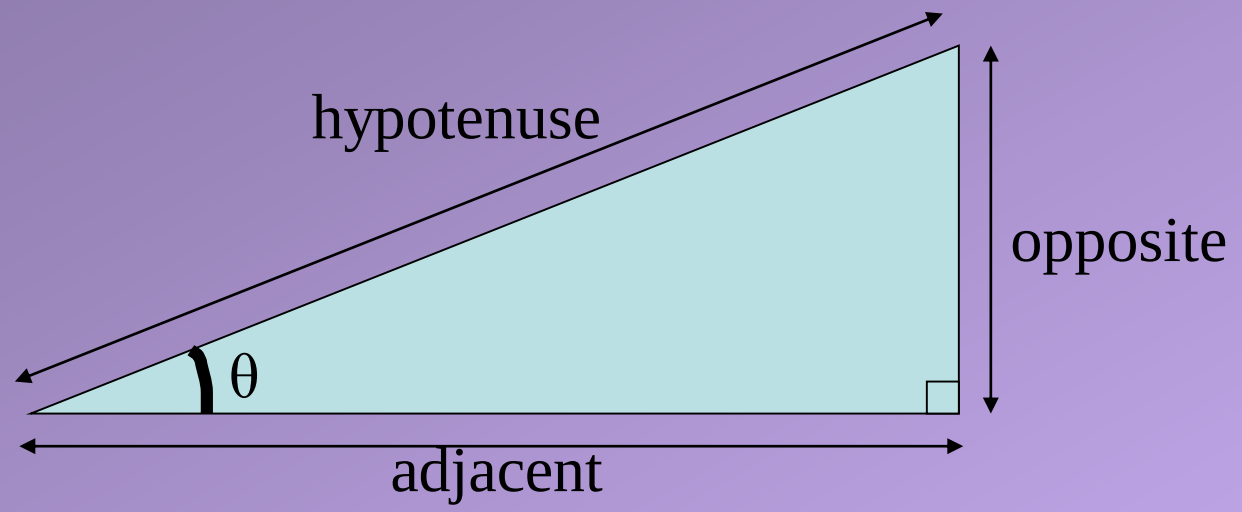


WARNING!

The angle <u>must</u> be given in <u>radians</u>!

Six Basic Trigonometric Functions

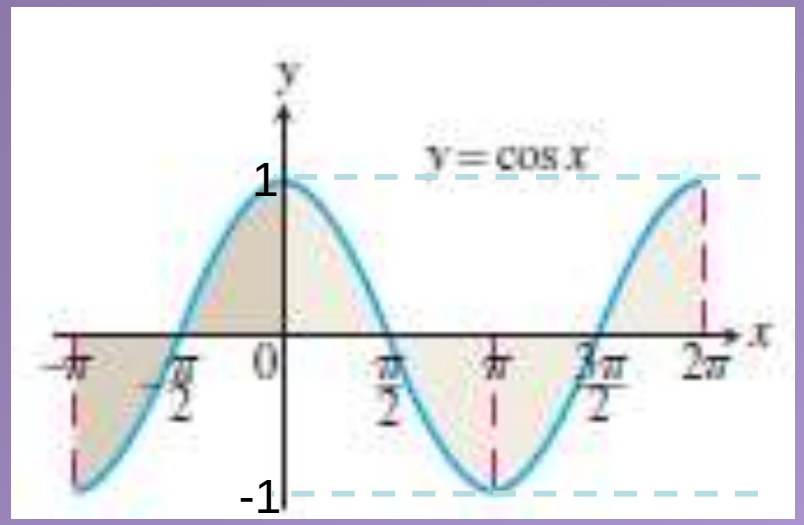
- Given a **right triangle**, and reference angle θ:

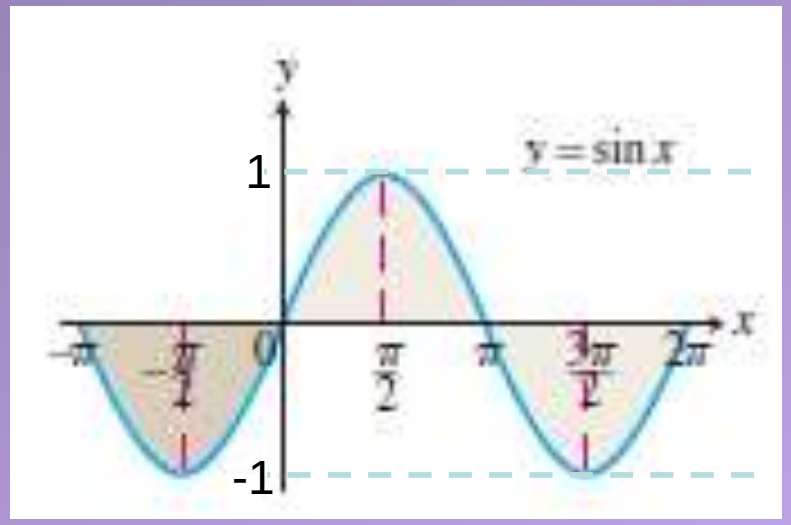


$$\sin\theta = \frac{opposite}{hypotenuse}, \qquad tan\,\theta = \frac{opposite}{adjacent}, \qquad \sec\theta = \frac{hypotenuse}{adjacent},$$

$$\cos\theta = \frac{adjacent}{hypotenuse}, \qquad \csc\theta = \frac{hypotenuse}{opposite}, \qquad \cot\theta = \frac{adjacent}{opposite}.$$

Sine and Cosine





Domain: $-\infty < x < \infty$

Range: $-1 < y < 1$

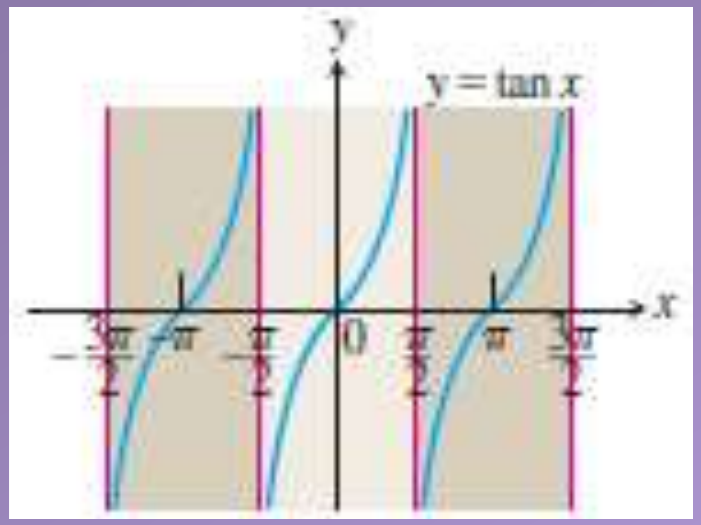
Period: 2π

Domain: $-\infty < x < \infty$

Range: $-1 < y < 1$

Period: 2π

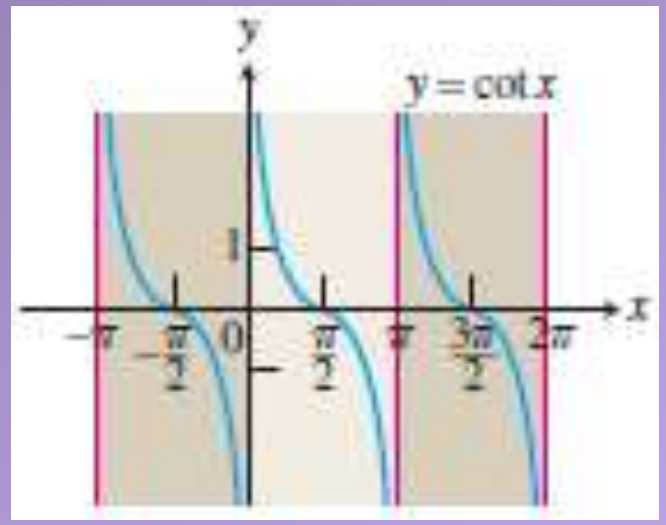
Tangent and Cotangent



Domain: $x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \cdots$

Range: $-\infty < y < \infty$

Period: π



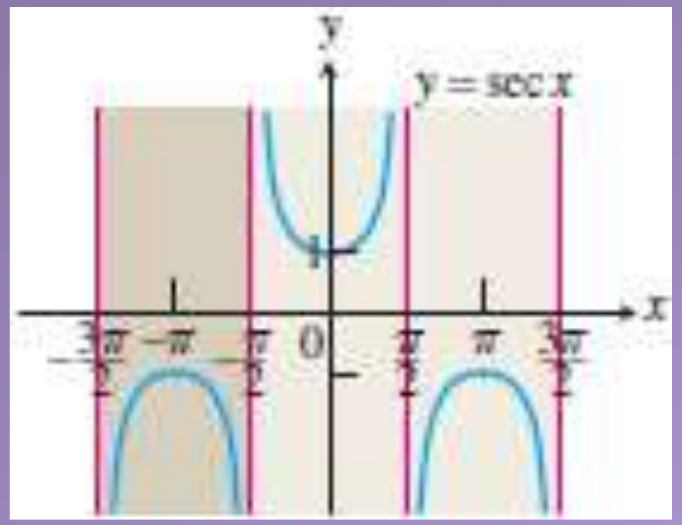
Remember, the cotangent is the reciprocal of the tangent.

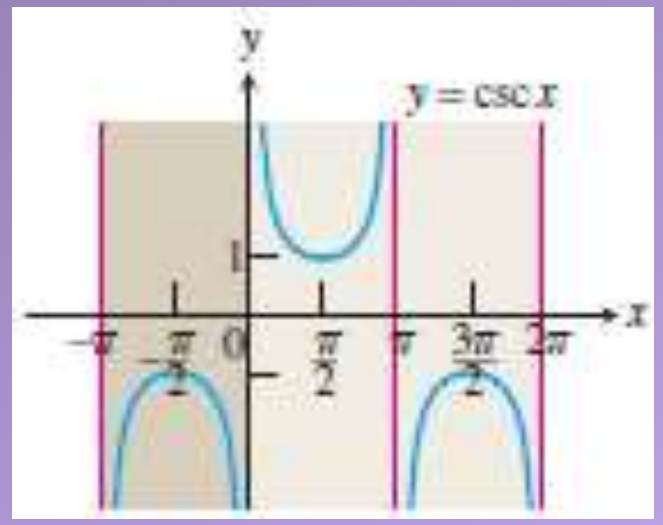
Domain: $x \neq 0, \pm\pi, \pm 2\pi, \cdots$

Range: $-\infty < y < \infty$

Period: π

Secant and Cosecant





Remember, the secant is the reciprocal of the cosine.

Remember, the cosecant is the reciprocal of the sine.

Domain: $x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \cdots$

Range: $y \leq -1$ and $y \geq 1$

Period: 2π

Domain: $x \neq 0, \pm\pi, \pm 2\pi, \cdots$

Range: $y \leq -1$ and $y \geq 1$

Period: 2π

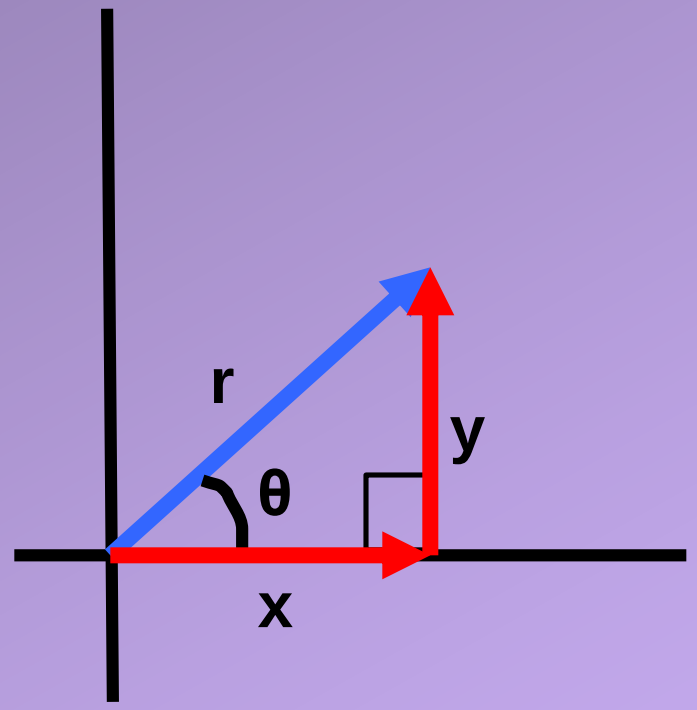
Most Common Application:

$$r = \sqrt{x^2 + y^2}$$

$$x = r\cos\theta$$

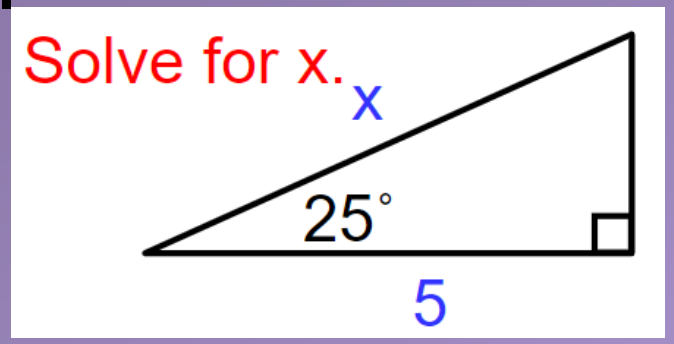
$$y = r\sin\theta$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$



- When we are trying to find a **side** we use sin, cos, or tan.
- When we are trying to find an **angle** we use inverse trigonometric functions: $\sin^{-1}$, $\cos^{-1}$, or $\tan^{-1}$.

Ex1.



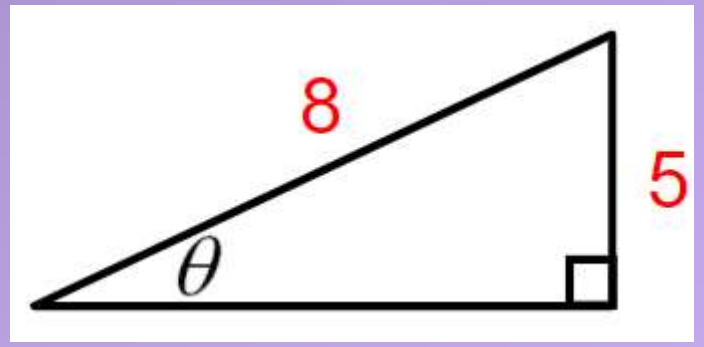
$$\cos 25° = \frac{5}{x}$$

$$x = \frac{5}{cos25°}$$

$$x \approx 5.5169$$

EX2.Sketch a triangle that has an acute angle θ, and then find θ.

$$\csc\theta = \frac{8}{5}$$



$$\sin\theta = \frac{5}{8}, \sin^{-1}\frac{5}{8} = \theta, \theta \approx 38.68°$$

Inverse Trigonometric Functions

$y = \sin^{-1} x$ or arcsin x

$y = \cos^{-1} x$ or arccos x

$y = \tan^{-1} x$ or arctan x

eg. arccos $(\frac{1}{2})$ → This question is actually asking: "what is the angle that has a cosine of $\frac{1}{2}$"? Ans: $\theta = 60°$

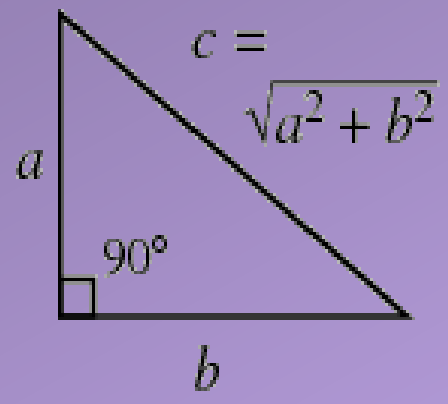
eg. Arctan(-8.45) $\approx -1.453001\, rad$

$\approx -83.25°$

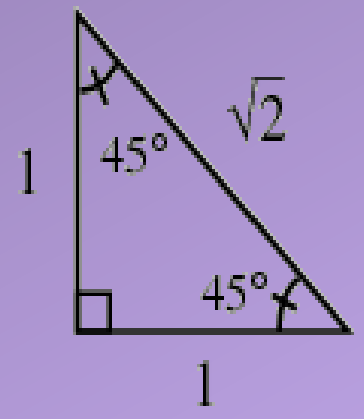
arctan$(-\frac{169}{20})$

Using the property where arctan$(-\frac{169}{20})$ = - arctan$(\frac{169}{20})$

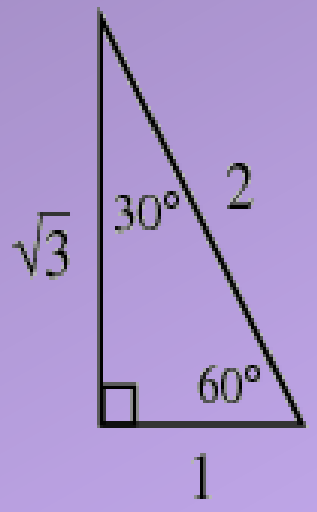
Common Triangles



general right triangle

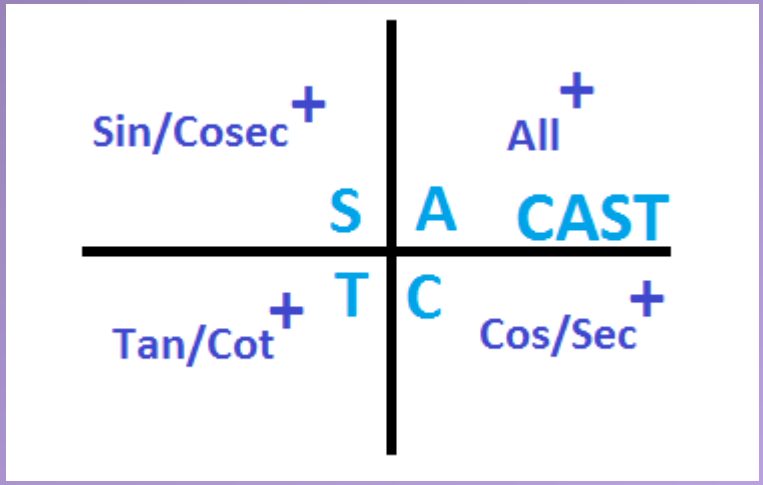


isosceles right triangle



30–60–90° triangle

CAST RULE



Remember the statement of “All Students Take Calculus”. It tells you which trigonometric functions are positive in each quadrant.

sin θ, cos θ and tan θ for selected values of θ

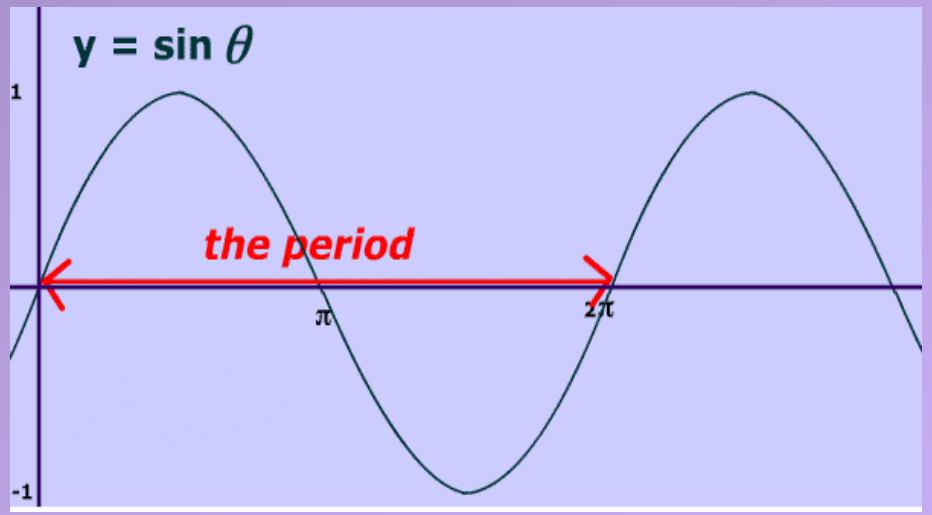
Angle θ				
Degrees	Radians	$\sin\theta$	$\cos\theta$	$\tan\theta$
0	0	0	1	0
30	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45	$\frac{\pi}{4}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1
60	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90	$\frac{\pi}{2}$	1	0	undefined
180	π	0	-1	0
270	$\frac{3\pi}{2}$	-1	0	undefined
360	2π	0	1	0

Periods of Trigonometric Functions

- *Periodic* function: A function f with a positive number p such that whenever θ is in the domain of f, so is $\theta + p$, and

$$f(\theta + p) = f(\theta)$$

- *(Fundamental) period* of f : smallest such number p, if it exists.



OR

A function $f(x)$ is **periodic** if there is a positive number p such that $f(x + p) = f(x)$ for every value of x. The smallest such value of p is the period of f.

In other words, periodic functions repeat themselves, and the period is the length of one cycle of the function.

- eg. The sine function repeats itself every 2π radians (or 360 degrees). The smallest number in which the function repeats itself is called the period. So, the sine function has a period of 2π. This means that the sine of $\frac{5\pi}{2}$ is the same as the sine of $\frac{\pi}{2}$.

Periods of Trigonometric Functions

- Periodic properties:

  $\sin(\theta + 2\pi) = \sin\theta$,

  $\cos(\theta + 2\pi) = \cos\theta$,

  $\csc(\theta + 2\pi) = \csc\theta$,

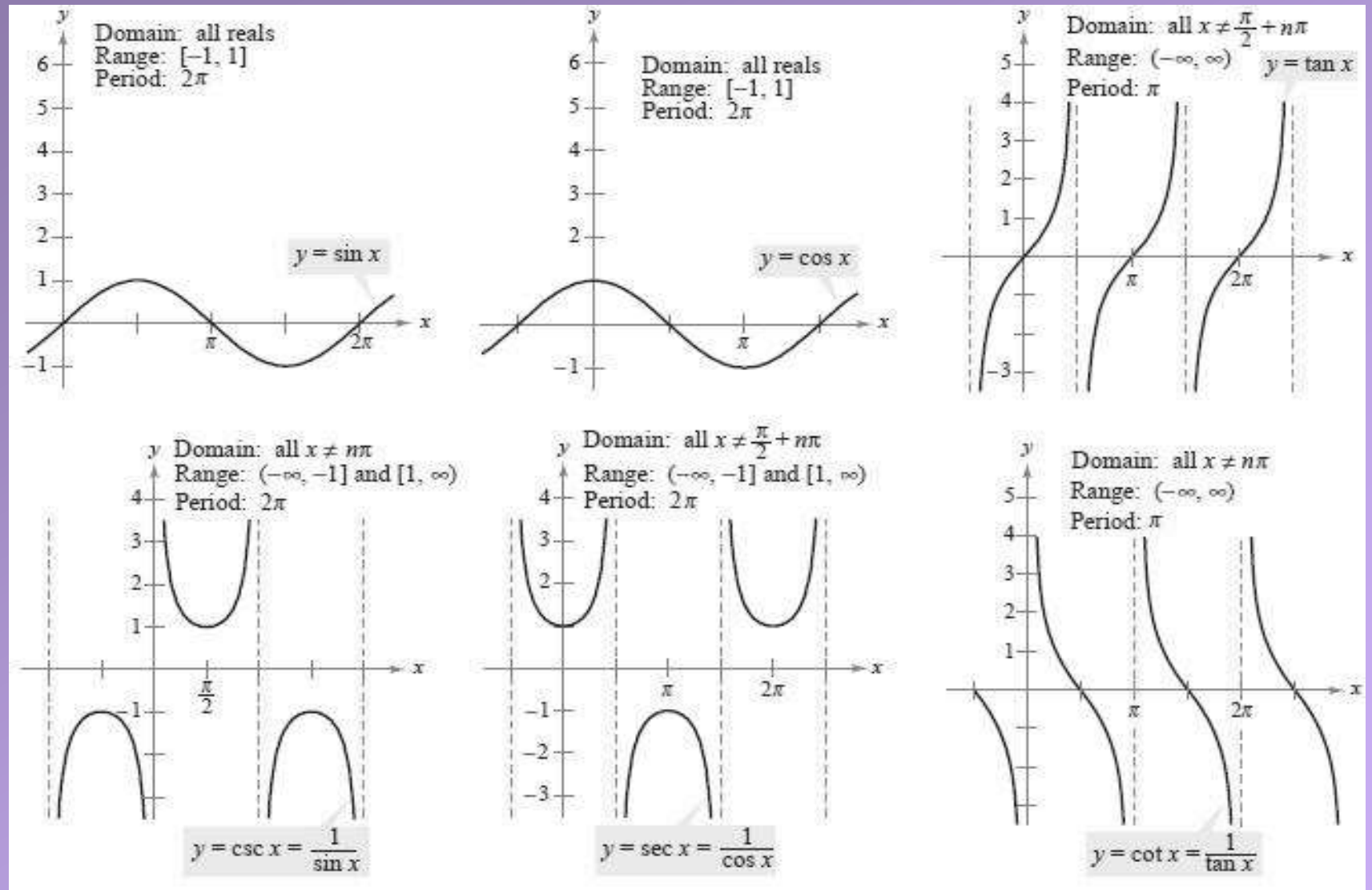
  $\sec(\theta + 2\pi) = \sec\theta$,

  $\tan(\theta + \pi) = \tan\theta$,

  $\cot(\theta + \pi) = \cot\theta$.

- <u>Sine, cosine, cosecant and secant</u> have period 2π; <u>tangent and cotangent</u> have period π.

Graphs of the six basic trigonometric functions using radian measure.



Periods of Trigonometric Functions

- Ex. Find the exact values of

(a) $\sin(7\pi) = \sin(\pi)$

(b) $\cos(37\pi/6) = \cos(\pi/6)$

(c) $\tan(19\pi/4) = \tan(3\pi/4)$

Even and Odd Trig Functions:

- “Even” functions behave like polynomials with even exponents, in that when you change the sign of x, the y value doesn’t change.
- Even functions are symmetric about the y - axis.
- “Odd” functions behave like polynomials with odd exponents, in that when you change the sign of x, the sign of the y value also changes.
- Odd functions have origin symmetry.

Even-Odd Properties

- A function f is even if $f(-\theta) = f(\theta)$ for all θ in the domain of f.
- A function f is odd if $f(-\theta) = -f(\theta)$ for all θ in the domain of f.
- <u>Theorem</u>. [Even-Odd Properties]

$\sin(-\theta) = -\sin(\theta)$

$\cos(-\theta) = \cos(\theta)$

$\tan(-\theta) = -\tan(\theta)$

$\csc(-\theta) = -\csc(\theta)$

$\sec(-\theta) = \sec(\theta)$

$\cot(-\theta) = -\cot(\theta)$

Sine is an **odd** function because: $\sin(-\theta) = -\sin(\theta)$

Cosecant, tangent and cotangent are also **odd**, because their formulas contain the sine function.

Cosine is an **even** function because: $\cos(-\theta) = \cos(\theta)$

Secant is also an **even** function, because it is the reciprocal of cosine.

Even-Odd Properties

- Cosine and secant are even functions.
- The other functions are odd functions.
- Ex. Find the exact values of

(a) $\sin(-30°)$ = -sin (30º) = …

(b) $\cos\left(-\frac{\pi}{4}\right)$ = cos(π/4) = …

(c) $\cot\left(-\frac{33\pi}{4}\right)$ = -cot(33 π/4) = -cot(π/4) = …

Fundamental Trigonometric Identities

- Quotient Identities

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

- Reciprocal Identities

$$\csc\theta = \frac{1}{\sin\theta} \qquad \sec\theta = \frac{1}{\cos\theta} \qquad \cot\theta = \frac{1}{\tan\theta}$$

- Pythagorean Identities

$$\sin^2\theta + \cos^2\theta = 1 \quad \tan^2\theta + 1 = \sec^2\theta \quad 1 + \cot^2\theta = \csc^2\theta$$

- Even-Odd Identities

$$\sin(-\theta) = -\sin\theta \qquad \cos(-\theta) = \cos\theta \qquad \tan(-\theta) = -\tan\theta$$
$$\csc(-\theta) = -\csc\theta \qquad \sec(-\theta) = \sec\theta \qquad \cot(-\theta) = -\cot\theta$$

More Formulas

- Angle Sum Formulas

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

- Double-Angle Formulas

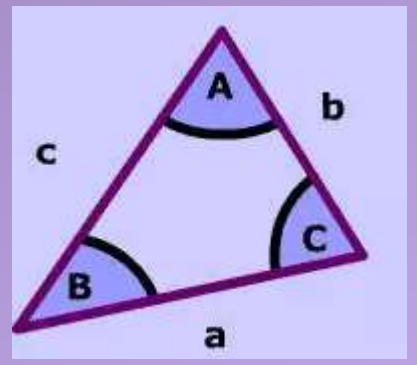
$$\sin 2\theta = 2\sin\theta\cos\theta$$

$$\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2 - 1 = 1 - 2\sin^2\theta$$

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta} = \frac{2}{\cot\theta - \tan\theta}$$

$$\cot 2\theta = \frac{\cot^2\theta - 1}{2\cot\theta} = \frac{\cot\theta - \tan\theta}{2}$$

More Formulas



- Additional Double-Angle Formulas

$$\sin^2 u = \frac{1-\cos(2u)}{2}$$

$$\cos^2 u = \frac{1+\cos(2u)}{2}$$

$$\tan^2 u = \frac{1-\cos(2u)}{1+\cos(2u)}$$

- What if it is not a right triangle?
  - Use the Law of Cosines

The Law of Cosines

In <u>*any*</u> triangle ABC, with sides a, b, and c,

$$a^2 = b^2 + c^2 - 2bc\cos A$$

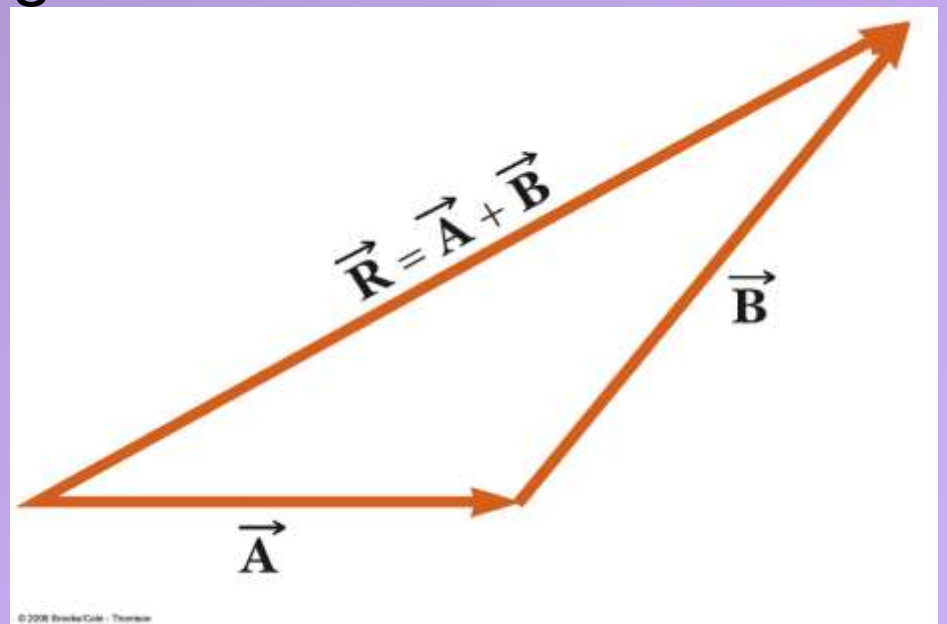
$$b^2 = a^2 + c^2 - 2ac\cos B$$

$$c^2 = a^2 + b^2 - 2ab\cos C.$$

More Formulas

- **Law of Cosines**
  - The square of the magnitude of the resultant vector is equal to the sum of the magnitude of the squares of the two vectors, minus two times the product of the magnitudes of the vectors, multiplied by the cosine of the angle between them.

$$R^2 = A^2 + B^2 - 2AB\cos\theta$$



1.5 EXPONENTIAL FUNCTIONS

Objectives

- To review exponential functions.
- To view their basic properties and uses.

Definition of Exponential Function

- The ***exponential function f*** with base *a* is defined by

$$f(x) = a^x$$

The reason a > 0 is that if it is negative, the function is undefined for -1 < x < 1.

where $a > 0$, $a \neq 1$, and x is any real number.

For instance,

$$f(x) = 3^x \text{ and } g(x) = 0.5^x$$

are exponential functions.

Examples

The value of $f(x) = 3^x$ when $x = 2$ is

$$f(2) = 3^2 = 9$$

The value of $f(x) = 3^x$ when $x = -2$ is

$$f(-2) = 3^{-2} = \frac{1}{9}$$

The value of $g(x) = 0.5^x$ when $x = 4$ is

$$g(4) = 0.5^4 = 0.0625$$

More…

- $-4^0 = 1$
- $\left(\frac{2}{3}\right)^{-4} = \left(\frac{3}{2}\right)^4$

- $y = a^x$ → eg. $y = 2^x$
  - Domain: $(-\infty, \infty)$
  - Range: $(0, \infty)$

- $f(x) = 2^{(x+2)} - 3$
  - Domain: $(-\infty, \infty)$
  - Range: $(-3, \infty)$

General Formula

- All exponential functions have the general format:

$$f(t) = A \cdot B^t$$

where

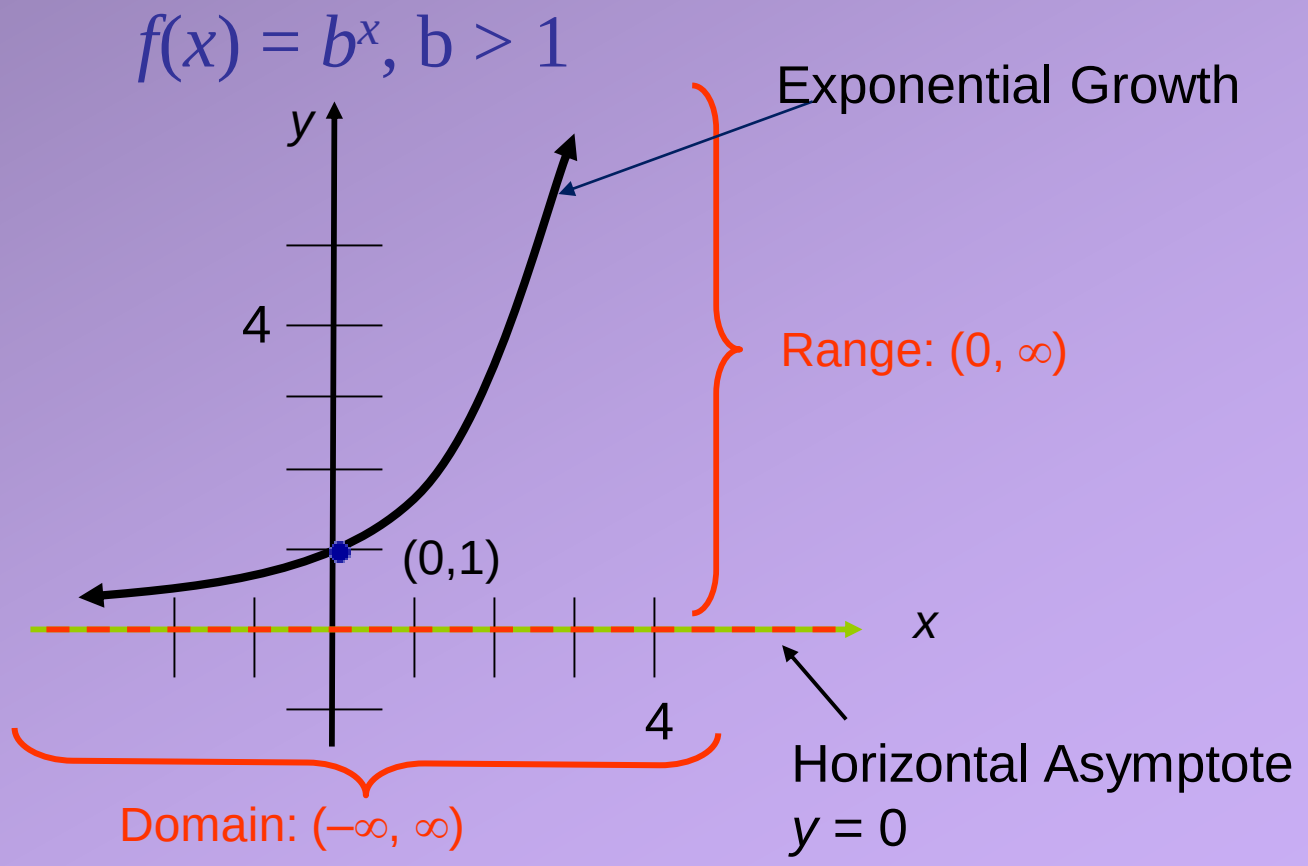
- A = initial value
- B = growth factor
- t = number of time periods

Characteristics of Exponential Functions

- The domain of $f(x) = b^{\boldsymbol{x}}$ consists of all real numbers. The range of $f(x) = b^{\boldsymbol{x}}$ consists of all positive real numbers.
- The graphs of all exponential functions pass through the point (0, 1) because $f(0) = b^0 = 1$.
- If $b > 1, f(x) = b^{\boldsymbol{x}}$ has a graph that goes up to the right and is an increasing function.
- If $0 < b < 1, f(x) = b^{\boldsymbol{x}}$ has a graph that goes down to the right and is a decreasing function.
- $f(x) = b^{\boldsymbol{x}}$ is a one-to-one function and has an inverse that is a function.

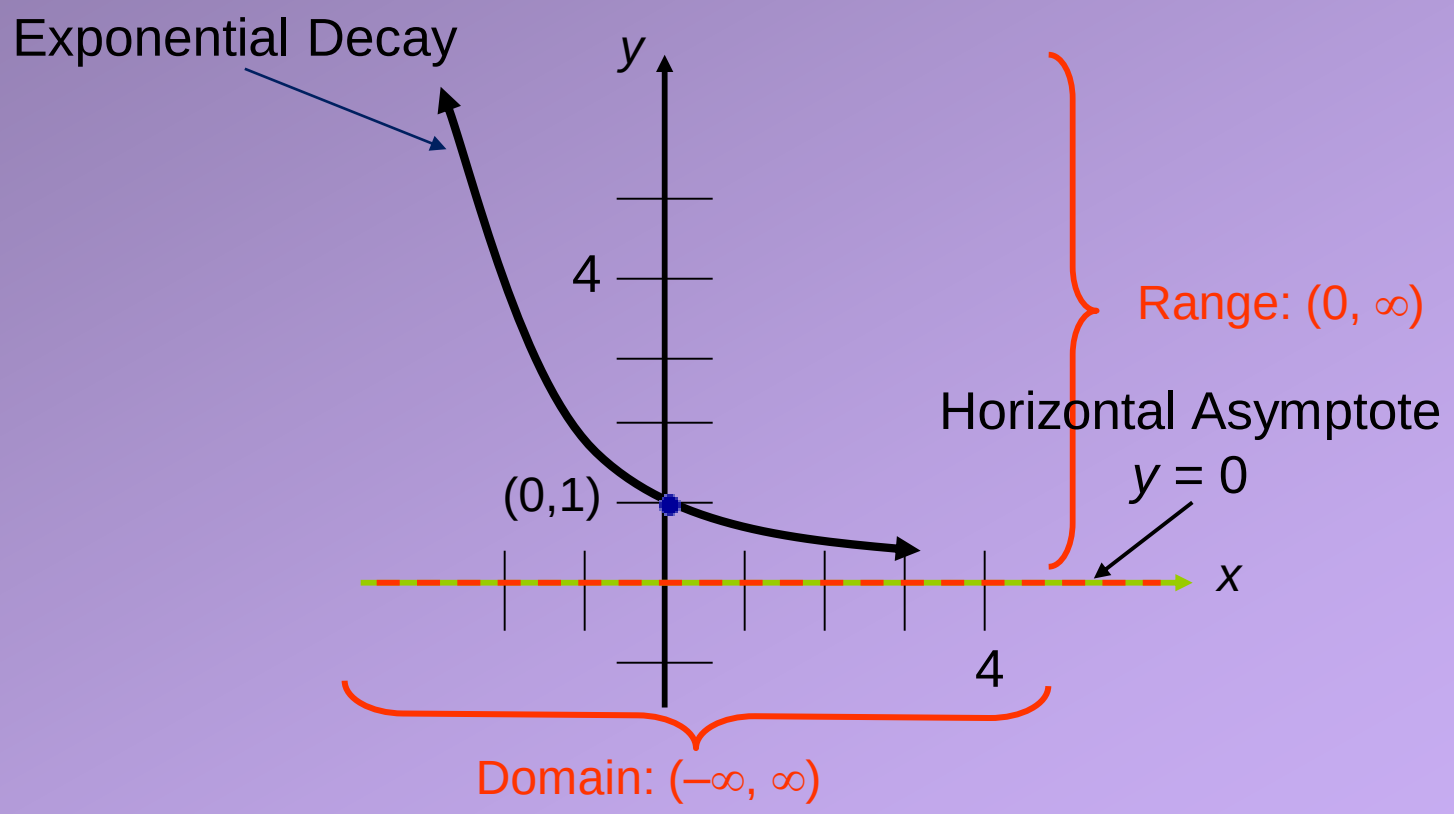
Characteristics of Exponential Functions

- The graph of $f(x) = b^x$ approaches but does not cross the x-axis. The x-axis is a horizontal asymptote.



Characteristics of Exponential Functions

$$f(x) = b^x,\ 0 < b < 1$$



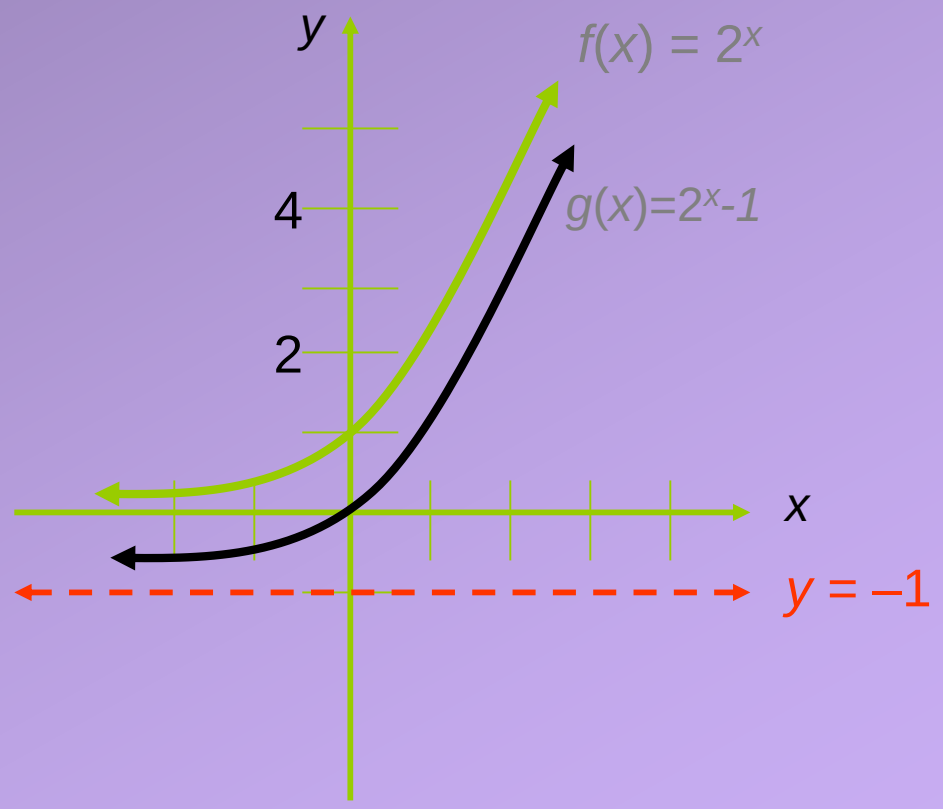
Transformations Involving Exponential Functions

Transformation	Equation	Description
Horizontal translation	$g(x) = b^{x+c}$	• Shifts the graph of $f(x) = b^x$ to the left c units if $c > 0$. • Shifts the graph of $f(x) = b^x$ to the right c units if $c < 0$.
Vertical stretching or shrinking	$g(x) = c\, b^x$	Multiplying y-coordintates of $f(x) = b^x$ by c, • Stretches the graph of $f(x) = b^x$ if $c > 1$. • Shrinks the graph of $f(x) = b^x$ if $0 < c < 1$.
Reflecting	$g(x) = -b^x$ $g(x) = b^{-x}$	• Reflects the graph of $f(x) = b^x$ about the x-axis. • Reflects the graph of $f(x) = b^x$ about the y-axis.
Vertical translation	$g(x) = -b^x + c$	• Shifts the graph of $f(x) = b^x$ upward c units if $c > 0$. • Shifts the graph of $f(x) = b^x$ downward c units if $c < 0$.

Transformations Involving Exponential Functions

Ex. Sketch the graph of $g(x) = 2^x - 1$. State the domain and range.

The graph of this function is a vertical translation of the graph of $f(x) = 2^x$ down one unit
.



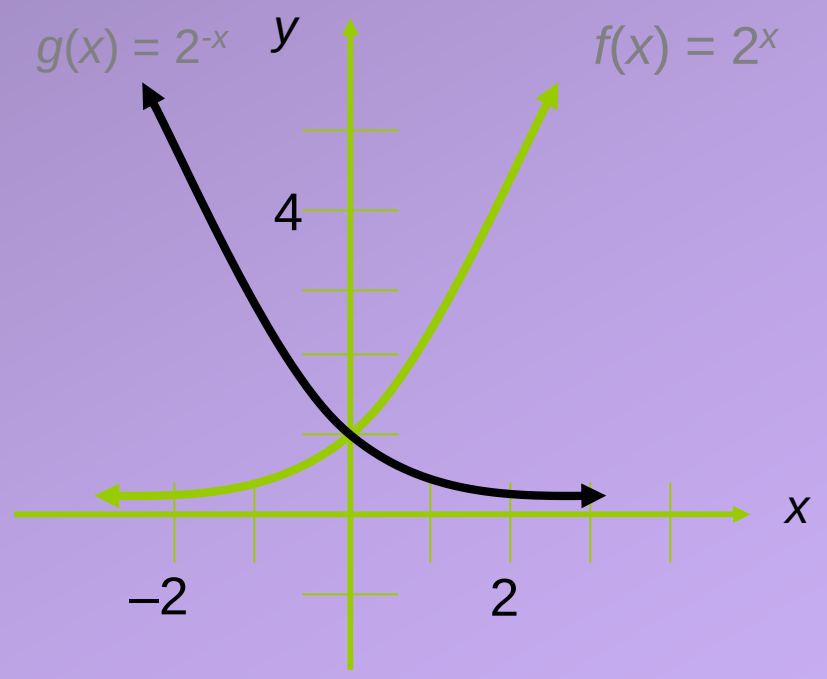
Domain: $(-\infty, \infty)$

Range: $(-1, \infty)$

Transformations Involving Exponential Functions

Ex. Sketch the graph of $g(x) = 2^{-x}$. State the domain and range.

The graph of this function is a reflection the graph of $f(x) = 2^x$ in the y-axis.



Domain: $(-\infty, \infty)$

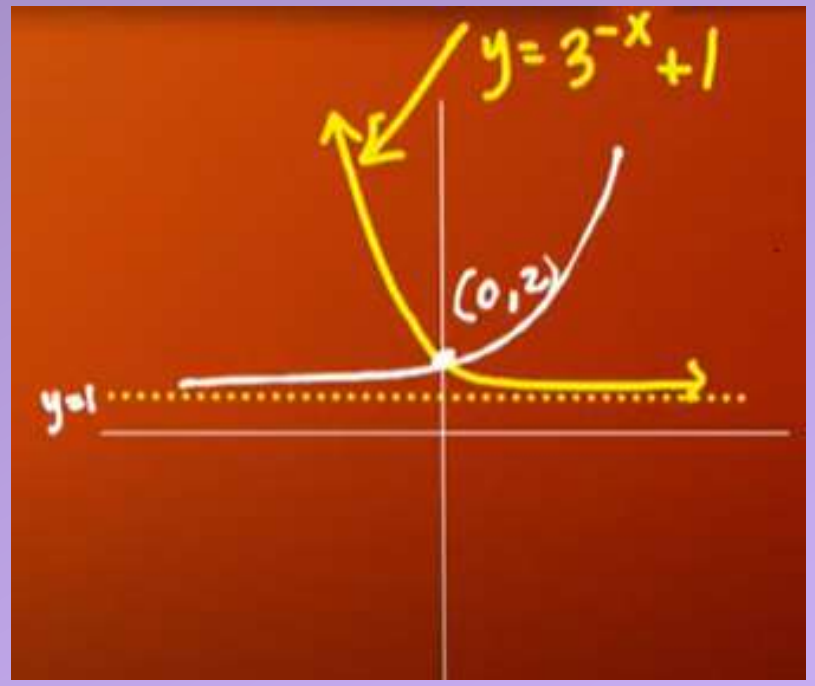
Range: $(0, \infty)$

Transformations Involving Exponential Functions

Ex. $y = 3^{-x} + 1$.

Up 1

Reflects over y



Horizontal asymptote: $y = 1$

y-intercept: (0, 2)

Domain: $(-\infty, \infty)$

Range: $(1, \infty)$

Transformations Involving Exponential Functions

Ex. $y = 3^{x-1} - 4$.

Right 1

down 4

Horizontal asymptote: $y = -4$

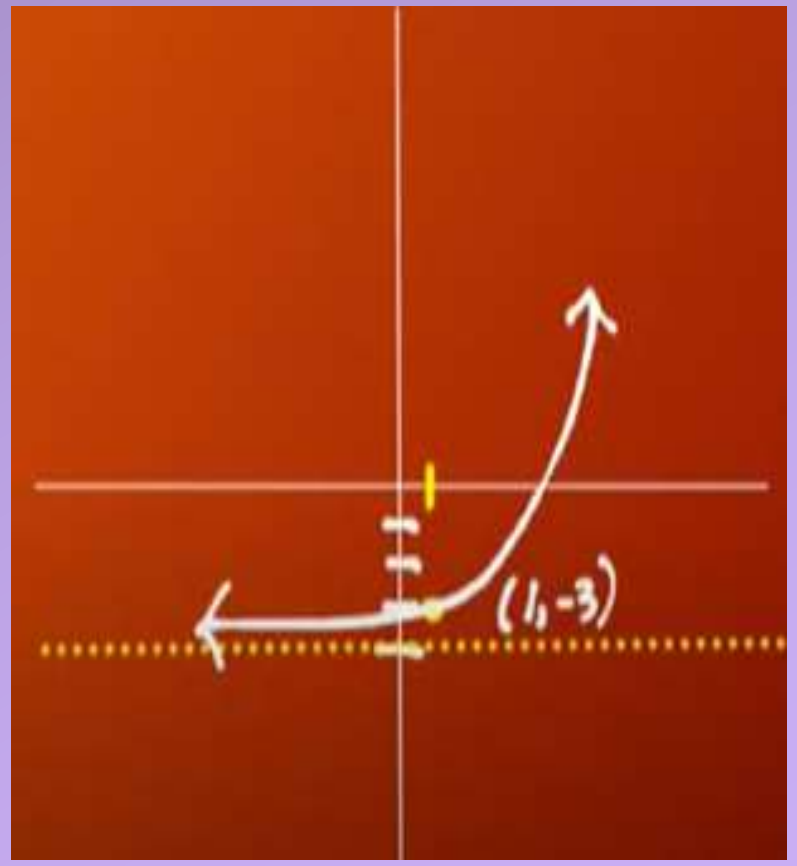
y-intercept: $(0, -3\frac{2}{3}) \longrightarrow$ x=0, $3^{0-1} - 4$

$= \frac{1}{3} - 4$

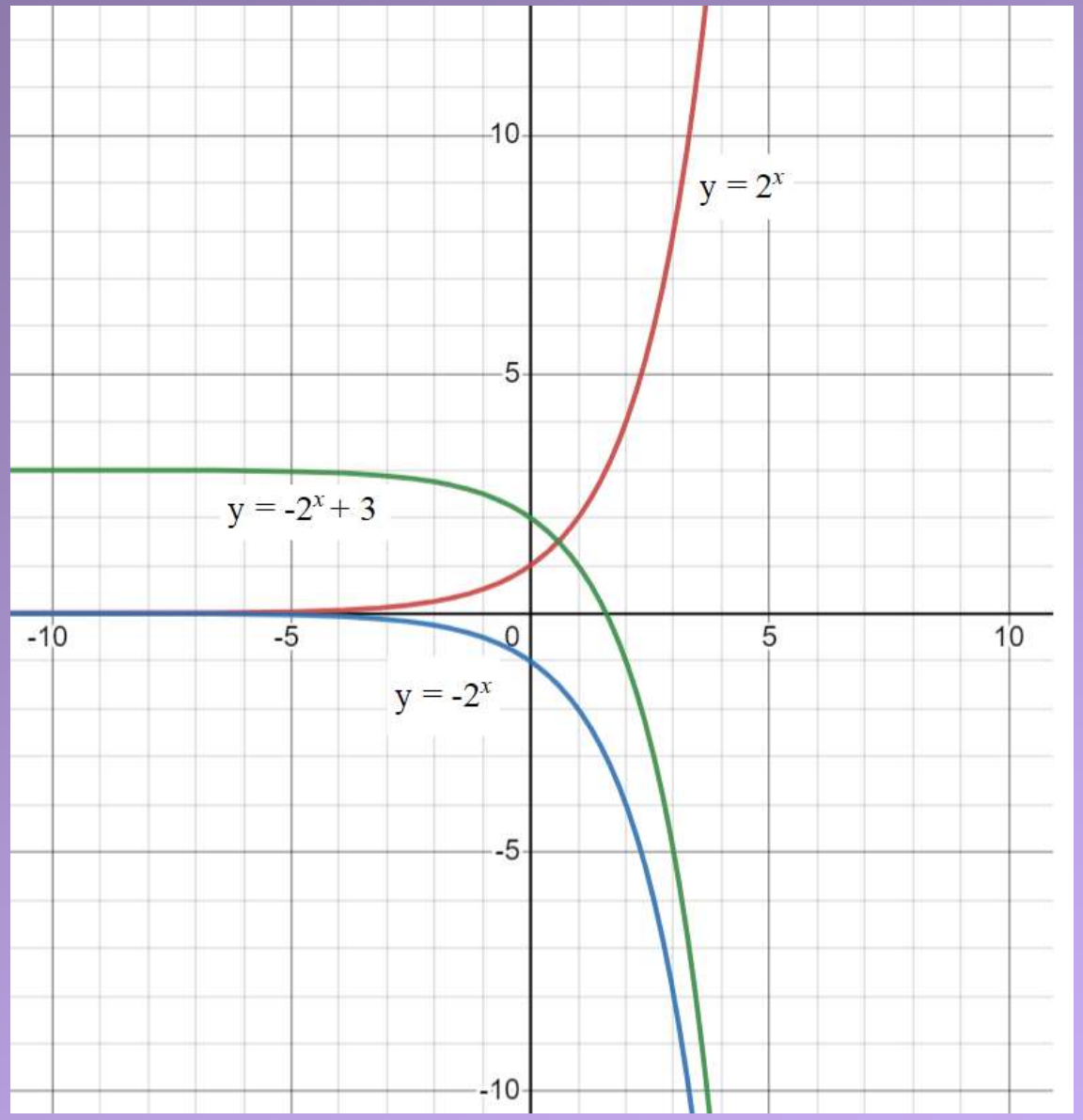
$= -3\frac{2}{3}$

Domain: $(-\infty, \infty)$

Range: $(-4, \infty)$



More…



Rules for Exponents

Exponent Rules

For $a > 0$, $b > 0$, x, y, *and n Real Numbers*

1. $a^x = a^y \iff x = y$
2. $a^x a^y = a^{x+y}$
3. $\frac{a^x}{a^y} = a^{x-y}$
4. $a^{-x} = \frac{1}{a^x}$
5. $(a^x)^y = a^{xy}$
6. $a^x b^x = (ab)^x$
7. $a^{\frac{x}{y}} = \sqrt[y]{a^x} = (\sqrt[y]{a})^x$ for x and y integers > 1
8. $a^0 = 1$
9. $a^1 = a$
10. $1^x = 1$

WVT

Rules for Exponents

Ex. Calculate a) $\frac{x^3}{x^2}$ b) 5^0 c) $(xy)^3$ d) x^{-2}

a) $\frac{x^3}{x^2} = x^{3-2} = x^1$

b) $5^0 = 1$

c) $(xy)^3 = x^3y^3$

d) $x^{-2} = \frac{1}{x^2}$

More examples…

- $-2^x = -\frac{1}{2}$

  $2^x = \frac{1}{2}$

So, x = -1.

- $8^x = 2$

  $(2^3)^x = 2^1$

  $3x = 1$

So, $x = \frac{1}{3}$.

The Natural Base ***e***

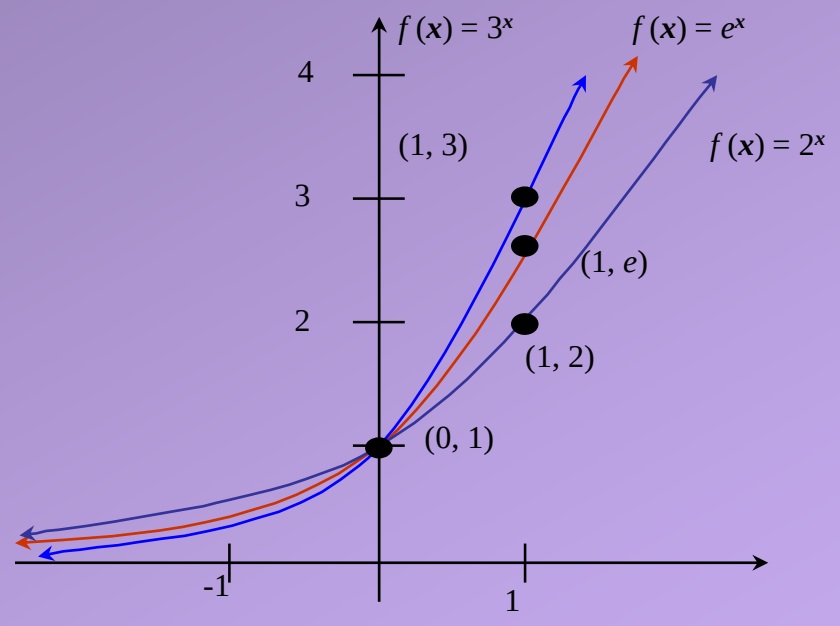
- An irrational number, symbolized by the letter *e*, appears as the base in many applied exponential functions. This irrational number is approximately equal to 2.72. More accurately, $e \approx 2.718281828\ldots$

- The number e is used in applications involving growth and decay; it is called as the natural base.
- Using techniques of calculus, it can be shown that

$$\left(1+\frac{1}{n}\right)^n \to e \ \textbf{as} \ n \to \infty$$

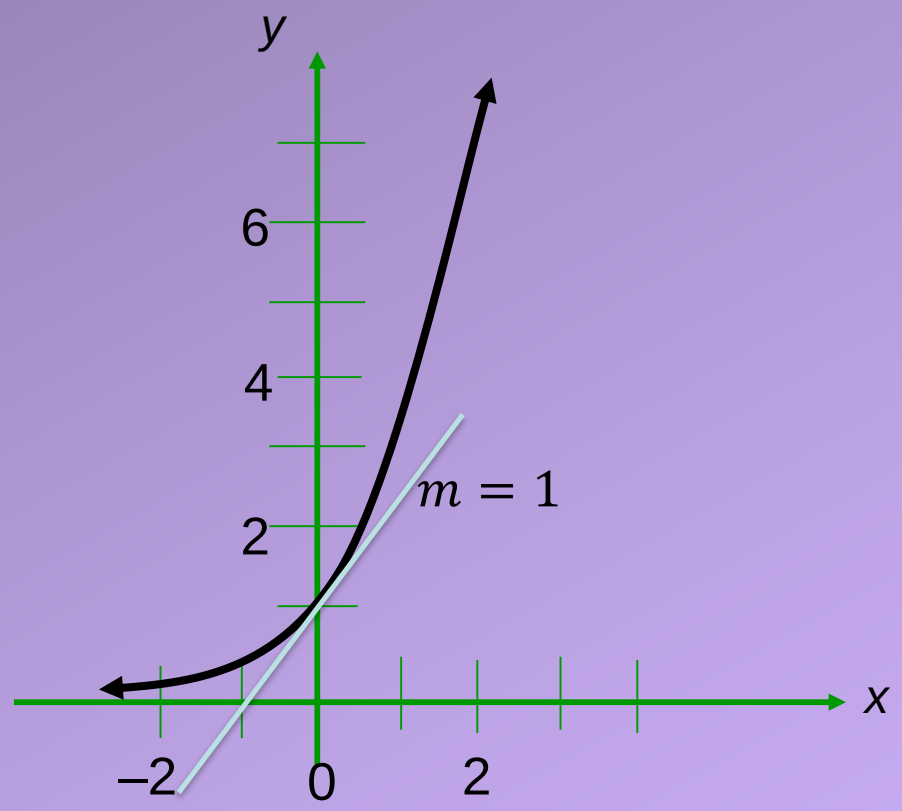
The Natural Exponential Function

- The function $f(\boldsymbol{x}) = e^{\boldsymbol{x}}$ is called the natural exponential function.



The Natural Exponential Function

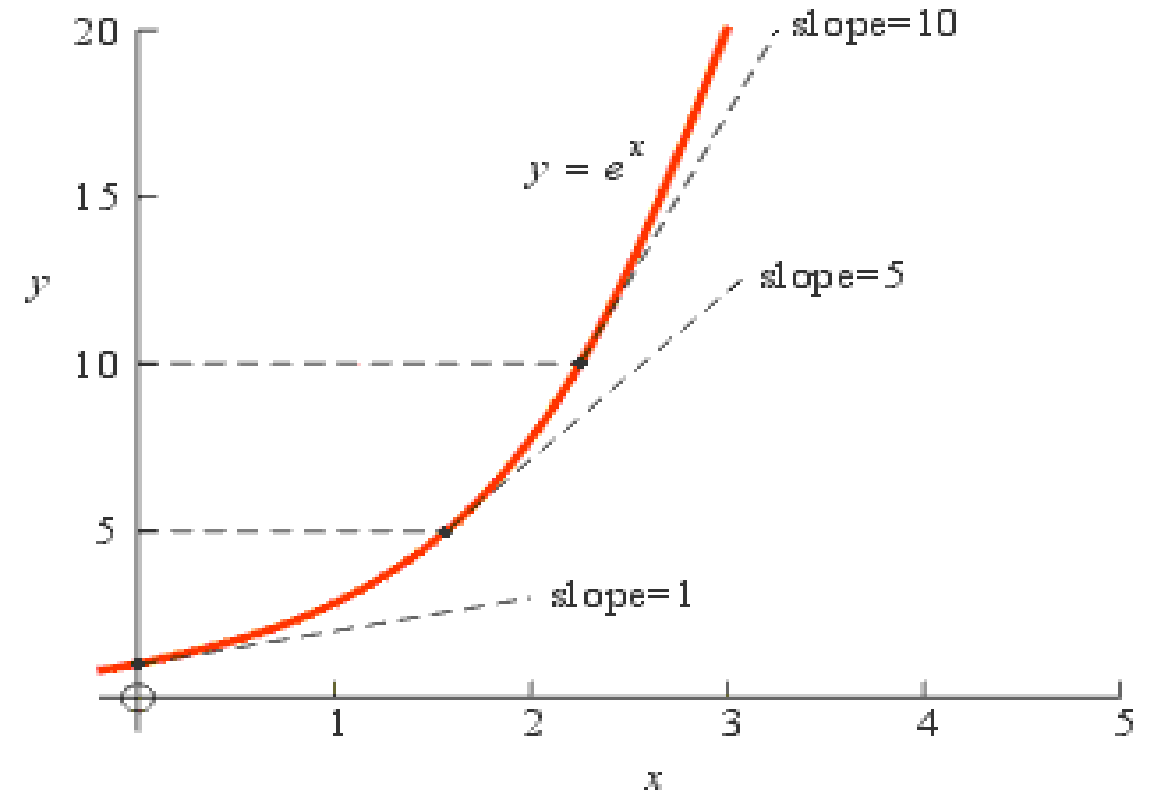
The graph of $f(x) = e^x$



x	$f(x)$
-2	0.14
-1	0.38
0	1
1	2.72
2	7.39

The Natural Exponential Function

- The slope is the same as the function value (the y-value) for all points on the graph.



Simple Interest $l = Prt$

- Calculate simple interest on the principal only, $l = Prt.$ Simple interest does not include the effect of compounding.
- P: Principal amount (the amount borrowed/invested)
- r: Rate of interest - given as a percent per year
- t: Time
- l: Interest amount

Simple Interest

The product of the principal, *P*, rate *r* and time *t* gives simple interest, *I*:

I = Prt

should be in the same time units such as months or years.

Compound Interest $A = P\left(1 + \frac{r}{n}\right)^{nt}$

- *P*: principal (amount of loan or deposit)
- *t*: length of investment/loan (in years)
- *r*: annual interest rate (as a decimal – not a percent)
- *n*: number of times interest is compounded in a year
- A: amount after *t* years

Formulas for Compound Interest

After t years, the balance, A, in an account with principal P and annual interest rate r (in decimal form) is given by the following formulas:

1. For n compoundings per year: $A = P\left(1+\frac{r}{n}\right)^{nt}$

   If P dollars is invested at a yearly rate of interest r per year, compounded n times per year after t years, the compound amount is $A = P\left(1+\frac{r}{n}\right)^{nt}$ dollars.

2. For Continuous Compounding Interest: $A = Pe^{rt}$

   If a deposit of P dollars is invested at a rate of interest r compounded continuously for t years, the compound amount is $A = Pe^{rt}$ dollars

Examples

- Inga Moffitt invests a bonus of RM9000 at 6% annual interest compounded (a) semiannually for 4 years (b) quarterly for 4 years (c) monthly for 4 years (d) daily for 4 years (e) continuously for 4 years. How much interest will she earn?

a) $A = P\left(1+\frac{r}{n}\right)^{nt}$

$$= 9000\left(1+\frac{0.06}{2}\right)^{2(4)}$$
$$= 9000(1.03)^{8}$$
$$= 11400.93$$

∴ RM11400.93 - RM9000 = RM2400.93.

b)

$$= 9000\left(1+\frac{0.06}{4}\right)^{4(4)}$$
$$= 9000(1.015)^{16}$$
$$= 11420.87$$

∴ RM11420.87 - RM9000 = RM2420.87.

(c) $9000\left(1+\frac{0.06}{12}\right)^{12(4)}$

$= 9000(1.005)^{48}$

$= 11434.40$

$\therefore$ RM11434.40 - RM9000

= RM2434.40.

(d) $9000\left(1+\frac{0.06}{365}\right)^{365(4)}$

$= 9000(1.000164)^{1460}$

$= 11441.02$

$\therefore$ RM11441.02 - RM9000

= RM2441.02.

- (e) $A = Pe^{rt}$

$9000e^{0.06(4)} = 9000e^{0.24}$

$=11441.24$

- If RM600 is invested in an account earning 2.75% compounded continuously, how much would be in the account after 5 years?

$$A = Pe^{rt}$$

$$A = 600e^{5(0.0275)} \approx 688.44$$

Example

Use $A = Pe^{rt}$ to solve the following problem: Find the accumulated value of an investment of $2000 for 8 years at an interest rate of 7% if the money is compounded continuously.

Solution

$A = Pe^{rt}$

$A = 2000e^{(.07)(8)}$

$A = 2000\ e^{(.56)}$

$A = 2000 * 1.75$

$A = \$3500$

Example: Choosing Between Investments

You want to invest $8000 for 6 years, and you have a choice between two accounts. The first pays 7% per year, compounded monthly. The second pays 6.85% per year, compounded continuously. Which is the better investment?

Solution The better investment is the one with the greater balance in the account after 6 years. Let's begin with the account with monthly compounding. We use the compound interest model with P = 8000, r = 7% = 0.07, n = 12 (monthly compounding, means 12 compoundings per year), and t = 6.

$$A = P\left(1+\frac{r}{n}\right)^{nt} = 8000\left(1+\frac{0.07}{12}\right)^{12*6} \approx 12{,}160.84$$

The balance in this account after 6 years is $12,160.84.

Example: Choosing Between Investments

You want to invest $8000 for 6 years, and you have a choice between two accounts. The first pays 7% per year, compounded monthly. The second pays 6.85% per year, compounded continuously. Which is the better investment?

Solution For the second investment option, we use the model for continuous compounding with P = 8000, r = 6.85% = 0.0685, and t = 6.

$$A = Pe^{rt} = 8000e^{0.0685(6)} \approx 12{,}066.60$$

The balance in this account after 6 years is $12,066.60, slightly less than the previous amount. Thus, the better investment is the 7% monthly compounding option.

Summary

- Functions – linear, polynomial, power function, etc.
- Even & Odd functions
- Combine functions
- Basic trigonometric functions
- Exponential functions